

Storm-Gauge Forward Model (MKM-SG-001)

Wolfram Oracle Benchmark of the Code and its v1.3 Document

MKM Research Labs

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1 Executive Summary

This document benchmarks the Storm-Gauge Forward Model (MKM-SG-001): the package `src/models/stormgauge/`, which turns a storm's track, intensity and footprint into a water level at each gauge. It asks whether the code computes what its model document (`storm_gauge.pdf`, v1.3 of 11 March 2026, as filed) says, and whether what the document says holds.

The code computes its documented equations. The haversine, the track interpolation, the three decay kernels, the intensity at the gauge and the power law agree with a 50-digit implementation of the document's equations to 9.4×10^{-15}. Over 780 runs of the full response (Eqs. 14–18), the water level agrees to 2×10^{-14} m, and every peak time and duration is identical (Table 1). Wolfram, computing each independently, agrees with the reference to 4.5×10^{-31}. That includes Wolfram's own implementation of the whole response, written separately from the reference.

The document's account of the model does not hold: 8 of 16 claims are met (Table 2). What holds is the arithmetic: all 58 figures in its four sensitivity tables reproduce, as do its kernel percentages, its calibration targets read as Eq. 13's steady state, and its timings. What does not hold is what the model is said to do, and where its output is said to go.

- **Nothing uses it.** Section 2.2 and Q5–Q6 say the model's responses fill `gaugets/` and feed the hazard curves, the flood risk model and MKM-TS-001. Nothing in `src`, `tools` or `scripts` imports the package. `gaugets/` is written by a scripted flood wave, not by this model.
- **A storm does not reach the levels the calibration names.** The targets of §3.5 (alert at $I \approx 40$, severe at 85+) hold only for a level held at Eq. 13's target indefinitely. A default storm crossing the gauge needs a peak of 56 to reach alert and 94 to reach warning, and does not reach severe at all.
- **The dynamics are not as described.** Peak level follows peak intensity by 3.5–5 h, not $\tau_{\text{lag}} = 2$ h. Doubling τ_{decay} lowers the peak by 19% and shortens the time above alert, rather than doubling it. The run ends with the storm, so durations and accumulation stop counting while the gauge is still high.
- **Two of its inputs are unread:** the intensity-to-level scale, whose Table 5 has six identical rows, and the storm's speed.
- **Two figures are overstated:** the effect of γ (§6.2), and the sphere's accuracy at UK scale, which is 0.32%, not under 0.3%.

Evidence. 8 checks, outcome: all pass (Table 1).

- Wolfram: 15.0.0 for Mac OS X ARM (64-bit) (May 26, 2026) via live kernel session (wolframclient).
- Code under test, at commit `7bbb63078a59`.
- Numerics: numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1.
- Generated 2026-11-04 by `python -m tools.wolfram_oracle sg001`.

2 Model Purpose and Scope

2.1 What is benchmarked

The package v1.3 documents, imported as its own tests import it:

- `forward_model.StormGaugeModel`: the haversine, the decay kernels, the intensity at a gauge, the power law, and `compute_response`, which produces the level series and its statistics;
- `storm_factory.create_storm`, which builds the storms the cases use;
- `data_structures`: the storm, gauge and response types.

2.2 What is not

- The calibration of γ, τ_{lag} and τ_{decay} to Environment Agency records (§4.4), and the backtest of §7.2, which are empirical. Nothing in the repository runs the model against gauge records, so neither can be reproduced from it.
- `find_nearest_track_point`, which `compute_response` does not call.
- The scripted wave that does write `gaugets/` (`port/rand/shared/gauge/gaugets_random`), and MKM-SS-001's sequence response, which re-implements Eq. 16 rather than calling this code.

This is a benchmark of the implementation (validation question Q8), not independent validation (Q9).

3 Mathematical Framework

The reference implements v1.3's equations directly from its text:

- the haversine of Eqs. 1–2, in its arctangent form, on a sphere of 6,371 km;
- the track interpolation of Eqs. 3–6, with α clamped to $[0, 1]$;
- the Gaussian, exponential and linear kernels of Eqs. 7–10, with $k = 1$ on the track;
- the intensity at the gauge, $I_s(t)\, k(\tilde{d}(t))$ (Eq. 11);
- the power law $\Delta L = (I/100)^{0.8} L_{\text{range}}\, s$ of Eqs. 12–13, with $L_{\text{high}} = 1.1 L_{\text{severe}}$ where a gauge has no historical high;
- the response of Eqs. 14–16 on §5.2's grid $t_i = i\,\Delta t$, $i = 0, \ldots, \lfloor T/\Delta t \rfloor$: the target is Eq. 13 at the intensity $\lfloor \tau_{\text{lag}}/\Delta t \rfloor$ steps earlier, approached at rate $\min(1, 3\Delta t/\tau_{\text{decay}})$ when rising and $\Delta t/\tau_{\text{decay}}$ when falling;
- the peaks, the durations of Eq. 17 and the accumulation of Eq. 18.

Where the document is silent, the reference assumes what the code does. Before the first track point the storm sits at it, and after the last at the last. Before τ_{lag} has elapsed, Eq. 15 reads the first step. The cases exercise both: one track starts late, and the Gaussian profile does not start at zero.

4 Benchmark Method

The cases.

- A default storm (Table 1: 24 h, 25 track points, footprint 40 km, triangular profile, Gaussian kernel at 0.5) crosses the Thames west to east, at peaks of 30, 50, 80 and 100.

- The same storm with every other kernel and profile, and an irregular storm whose track starts late, turns and ends early.
- Five gauges: on the track, 0.1, 0.2 and 0.4° north of it, and one with a historical high and a sensitivity of 1.3.
- Four settings of $(\Delta t, \tau_{\mathrm{lag}}, \tau_{\mathrm{decay}})$, one with a lag that is not a whole number of steps; and durations of 24, 23.7 and 48 h.

The closed-form families also run on distances and intensities below zero, at zero, and far beyond the footprint.

The printed tables. Tables 4 and 5 were produced by the house sensitivity generator; each figure is judged as the code's value, rounded and formatted as that generator does. Tables 6 and 7 were written by hand, and are judged to the places they print.

The statements. Each statement about the model's behaviour is judged by running the code, which the checks show computes the document's equations.

- The calibration targets are solved twice: from Eq. 13 directly, and as the least peak intensity at which a default storm crossing the gauge reaches each stage.
- The lag is measured over every case in which the gauge feels the storm.
- τ_{decay} and τ_{lag} are doubled on the peak-100 crossing.
- The durations are compared with the same storm run on to 72 h.
- An input is unread if changing it changes no response.
- Where the output goes is found by searching the tree for anything that imports the package.

Wolfram. Wolfram works from each input's exact value, independently of the reference:

- the great-circle distance, as R times the angle between the two points' unit vectors (not the haversine);
- the kernels and the power law;
- the whole response, written again in the Wolfram Language for three storms (the default crossing, an exponential storm with a beta profile off the track, and the irregular storm), returning the peak level, the accumulation and the final level;
- the true distance between six UK pairs, by `GeoDistance` on WGS84.

5 Validation and Backtesting

Table 1: Benchmark checks

Check	Result	Limit	Outcome
Eqs. 1–2, the haversine: code against the reference, worst error (m)	9.0×10^{-10}	1.0×10^{-6}	pass
Eqs. 7–10, the decay kernels: code against the reference, worst relative error	9.8×10^{-16}	1.0×10^{-12}	pass
Eqs. 3–6 and 11, the intensity at the gauge: code against the reference, worst relative error	9.4×10^{-15}	1.0×10^{-12}	pass
Eqs. 12–13, the intensity-to-level power law: code against the reference, worst relative error	8.1×10^{-16}	1.0×10^{-12}	pass
Eqs. 14–16, the response over 780 runs: worst level error (m)	1.7×10^{-14}	1.0×10^{-6}	pass
§3.7, peak level, peak intensity and accumulation (Eq. 18): worst relative error	5.8×10^{-14}	1.0×10^{-12}	pass
§3.7, peak times and durations (Eq. 17): statistics that differ	0	0	pass
Wolfram agrees with the reference	4.5×10^{-31}	1.0×10^{-25}	pass

5.1 The model document's claims

Table 2: The model document's claims, tested

Claim	Stated	Benchmark finds	Outcome
§3.3.1	Gaussian: 60.7% of on-track intensity at $\tilde{d} = \sigma$, 13.5% at 2σ	as printed	met
Table 4	decay by kernel and distance	21 of 21 figures reproduced	met
Table 5	contribution by intensity and scale factor	30 of 30 figures reproduced	met
Table 6	$k(1.0)$ for each kernel	3 of 3 figures reproduced	met
Table 7	$(1/2)^\gamma$ for four exponents	4 of 4 figures reproduced	met
Tables 1, 5	the scale factor and the storm's speed are model inputs	changing either changes nothing; Table 5's six rows are identical	**not met**
§6.2	±25% in γ moves a mid-range contribution by 20–25%	+14.9% and -12.9% at $I = 50$; 9.7%–20.1% over $I = 40$–60	**not met**
§3.5	the level reaches alert at $I \approx 40$, warning at 60–70, severe at 85+, the high at 100	Eq. 13's target does: at 38.7, 64.7, 86.0 and 100	met
§3.5	. . . and so does a storm of that intensity	a default storm crossing the gauge reaches alert at peak 56.0, warning at 93.6, severe never, even at 100	**not met**
§3.7, §7.1	peak level follows peak intensity by about $\tau_{\text{lag}} = 2$ h	by 3.5 to 5.0 h	**not met**
§6.3	doubling τ_{decay} roughly doubles the time above alert; peak barely moves	12.0 h becomes 9.5 h; the peak moves -18.8%	**not met**
§6.3	a longer τ_{lag} moves the timing, not the peak	doubled, the peak moves +0.0% and +2.0 h	met
§3.7	durations are the total time above each stage	the run stops at the storm's end, the level still 2.84 m up (68.7% of its rise); run on, 12.0 h above alert becomes 13.0 h	**not met**
§8(5)	the sphere is within 0.3% of the ellipsoid at UK scale	-0.32% to -0.06% against `GeoDistance` on WGS84	**not met**
Q3	under 10 ms a gauge; 25 gauges under 250 ms	0.3 ms; 6 ms	met
§2.2, Q5, Q6	its responses fill `gaugets/` and feed the hazard curves, MKM-PF-001 and MKM-TS-001	nothing in `src`, `tools` or `scripts` imports it	**not met**

5.2 What the findings mean

The model is not on any production path. The document places it at the centre of the chain: its responses stored in `gaugets/`, consumed by the hazard curves, the flood risk model and the time-series statistics (§2.2, Q5, Q6). No module imports `models.stormgauge`. `gaugets/` is written by `port/src/gauge/gaugets.py` from `gaugets_random`, a scripted wave with staggered peaks and a sinusoidal ripple. So the governance record describes as production a model that produces nothing. The code is correct; the record of where it is used is not.

The calibration describes a steady state no storm reaches. Eq. 16 closes only $3\Delta t/\tau_{\text{decay}} = 12.5\%$ of the gap to its target each half-hour. A triangular storm's target falls again long before the level arrives: the peak-100 crossing rises to 74% of its target. The intensity bands of §3.5 are therefore Eq. 13's, not the model's. A storm must be far stronger than the bands say to reach each stage.

The same smoothing explains the lag, τ_{decay} and the durations.

- The level keeps rising for as long as the falling target is still above it. That adds 1.5–3 h to the 2 h lag. No unit test checks the lag, although §7.1 says one does.
- τ_{decay} sets the rise as well as the fall. Doubling it halves the rising rate, and so lowers the peak, before it lengthens anything.
- The grid stops at T (§5.2). At the end of the peak-100 crossing the gauge is still 2.84 m above base, 69% of its rise, and above alert. Run on, it stays above alert another hour.

The smaller findings.

- `intensity_to_level_scale` is stored and never read; Table 5's caption presents it as a sensitivity, and its rows are identical. The storm's `speed_kmh` is computed and never read.
- A ±25% change in γ moves the contribution at $I = 50$ by +14.9% and −12.9%, and by 10–20% across $I = 40$–60, not 20–25%.
- The sphere understates the default storm's own east–west track by 0.32%, slightly more than §8(5)'s 0.3%; north–south it errs by 0.06%. That is immaterial beside the findings above.

6 Model Limitations and Known Weaknesses

1. **Benchmark, not validation.** The code is shown to compute its documented equations. Whether $\gamma = 0.8$ and the time constants fit UK catchments is an empirical question, and nothing in the repository tests it.

2. **The claims are judged on the code's defaults.** Every behavioural claim is judged on Table 1's default storm and Table 2's default gauge. A gauge with its own historical high moves the steady-state bands. It does not change the smoothing that keeps a storm from reaching them.

7 Document History

Date	Version	Description	Author
04-Nov-2026	0.1	First issue: tests the filed v1.3	David K Kelly